

# VARUN RANA

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Dehradun, India

## Profile Summary

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Motivated Computer Science undergraduate with hands-on experience in MERN stack and real-time applications using technologies like React, Node.js, WebRTC, and Socket.io. Focused on building practical projects and seeking opportunities to apply and grow development skills in real-world.

## Education

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### Graphic Era Hill University

*B.Tech in Computer Science & Engineering*

Dehradun, Uttarakhand

2023 – 2027 — GPA: 7.9

### Doon Public School, Bhaniyawala

*Higher Secondary (12th), ISC*

2022 – 2023 — 79%

## Technical Skills

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- **Languages:** C, C++, JavaScript
- **Frontend:** React.js, HTML5, CSS3, Tailwind CSS, EJS
- **Backend:** Node.js, Express.js, REST APIs, Socket.io, WebRTC
- **Databases:** MongoDB, Mongoose, MySQL, PostgreSQL, Appwrite
- **ML/AI:** PyTorch, CycleGAN, Deep Learning
- **Tools & Platforms:** Git, GitHub, Cloudinary, Render, VS Code, Passport.js

## Projects

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### BookMyStay | *Node.js, Express.js, MongoDB, EJS, Cloudinary, Passport.js* | Live Demo

- Built a full-stack rental listing platform enabling users to browse, create, edit, and delete property listings with location-based search across 50+ global stays.
- Implemented secure user authentication and session management using Passport.js with role-based access control, only listing owners can modify or remove their content.
- Integrated Cloudinary for image upload and storage, and embedded interactive maps on each listing page for geolocation-based discovery.
- Enforced server-side schema validation on all form inputs; deployed on Render with MongoDB Atlas as the production database.

### Voxa | *React.js, Node.js, WebRTC, Socket.io* | Live Demo

- Developed a real-time video conferencing application with peer-to-peer (P2P) connections using WebRTC, supporting HD video, audio, and screen sharing.
- Implemented live chat alongside video using Socket.io for bidirectional event communication; architected signaling server to manage WebRTC handshakes and ICE negotiation.
- Engineered secure, direct P2P connections to minimize latency and reduce server load, achieving sub-200ms connection establishment in local testing.

### Echo → MRI Translation (CycleGAN) | *Python, PyTorch, CycleGAN, Deep Learning* | Live Demo

- Trained a CycleGAN model for unpaired image-to-image translation from echocardiography (ultrasound) to MRI-like cardiac images, addressing the diagnostic quality gap in rural healthcare.
- Implemented dual generator-discriminator architecture with cycle-consistency loss to preserve cardiac structure and anatomical features during domain translation.
- Eliminated the need for paired medical datasets — a critical constraint in real-world clinical data settings making the solution viable for deployment in resource-limited environments.

## Achievements & Certifications

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- Solved 150+ DSA problems on LeetCode/GeeksforGeeks